QF2104 Tutorial 9

Question 1

A financial market consists of four risky assets and a risk-free asset. The covariance matrix and expected rates of return of the risky assets are given respectively by

$$C = \begin{pmatrix} 1 & 1 & 0 & 0 \\ 1 & 3 & \alpha & 1 \\ 0 & \alpha & 2 & 1 \\ 0 & 1 & 1 & 4 \end{pmatrix}, \qquad \mu = \begin{pmatrix} 1 \\ 3 \\ 2 \\ 3 \end{pmatrix}$$

C is the covariance between r_i and r_j. It is given further that $\mu^T C^{-1} \mu = 35/3$, and $C^{-1}1 = (2/3, 1/3, 2/3, 0)^T$

i) Find the exact value of α

Answer: Notice $CC^{-1}1 = 1$. Hence

$$\begin{pmatrix} 1 & 1 & 0 & 0 \\ 1 & 3 & \alpha & 1 \\ 0 & \alpha & 2 & 1 \\ 0 & 1 & 1 & 4 \end{pmatrix} \begin{pmatrix} 2/3 \\ 1/3 \\ 2/3 \\ 0 \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \\ 1 \\ 1 \end{pmatrix}$$

Taking the second row

$$2/3 + 3(1/3) + \alpha(2/3) + 1(0) = 1$$

Hence $\alpha = -1$

ii) An economist seeks the minimum-variance portfolio whose weight vector w is of the form $(\beta, \beta, 1 - 3\beta, \beta)^T$. Find the weight vector of this portfolio.

Answer: from first principles lets find β whose weight vector gives min. σ_p^2

$$\sigma_p^2 = wCw = \begin{pmatrix} \beta & \beta & 1-3\beta & \beta \end{pmatrix} \begin{pmatrix} 1 & 1 & 0 & 0 \\ 1 & 3 & -1 & 1 \\ 0 & -1 & 2 & 1 \\ 0 & 1 & 1 & 4 \end{pmatrix} \begin{pmatrix} \beta \\ \beta \\ 1-3\beta \\ \beta \end{pmatrix} = 30\beta^2 - 12\beta + 2$$

$$\frac{d\sigma_p^2}{d\beta} = 60\beta - 12 = 0$$

Hence $\beta = 1/5, \sigma_p^2 = 4/5$

iii)Find the equation of the minimum-variance frontier of risky assets, giving your answer in the form$\sigma^2 = A(\mu - \mu_g)^2 + \sigma_g^2$, where A is a constant, (σ, μ) is any point on the frontier and is (σ_g, μ_g) the point corresponding to the global minimum-variance portfolio (GMVP) of risky assets.

Answer: Recall our MVP formula in terms of σ

$$\sigma_p^2 = \frac{a\mu_p^2 - 2b\mu_p + c}{ac - b^2}$$

Thus with $a = 5/3, b = 3, c = 35/3$, we have

$$\sigma_p^2 = \frac{5/3\mu_p^2 - 6\mu_p + 35/3}{175/9 - 9} = \frac{15}{94}\left(\mu - \frac{9}{5}\right)^2 + \frac{3}{5}$$

iv)The equation of the Capital Market Line is given by $\mu = \frac{79}{5}\sqrt{10/237}\sigma + 1/5$ write down the value of r_f and find the mean rate of return of the tangency portfolio.

Answer: The risk free rate $r_f = 1/5$ which is the constant term in the CML equation

Question 2

A financial market consists of three risky assets and a risk-free asset. Portfolios

$$g, m, d$$

are respectively the global minimum-variance portfolio of risky assets, the market (tangency) portfolio and a minimum-variance portfolio which has the same variance as the market portfolio. The portfolio d invests in risky assets only. The risk-free asset has rate of return r_f. The covariance matrix C and mean value vector μ is given such that

$$C^{-1}1 = \begin{pmatrix} 2 \\ 1 \\ 5 \end{pmatrix}, \quad C^{-1}\mu = \begin{pmatrix} 3/2 \\ 4 \\ 9/2 \end{pmatrix}, \quad \mu^T C^{-1}\mu = 59/4$$

The tangency portfolio mean return is $\mu_T = 13/8$.

$$a = 1^T C^{-1}1 = 8, \quad b = 1^T C^{-1}\mu = 10, \quad c = \mu^T C^{-1}\mu = 59/4$$

i)Find the weight vector, mean and variance of portfolio g

$$w_g = \frac{C^{-1}1}{1^T C^{-1}1}, \quad \mu_g = b/a, \quad \sigma_g^2 = 1/a$$

Answer: $w_g = (1/4, 1/8, 5/8)^T, \quad \mu_g = 5/4, \quad \sigma_g^2 = 1/8$

ii) Find the equation of the minimum variance frontier for risky assets only. Give your answer in the form $\sigma^2 = A(\mu - B)^2 + C$

Answer: Use question 1 part 3 concepts

$$\sigma^2 = A(\mu - \mu_g)^2 + \sigma_g^2$$

Inspecting the close form

$$\sigma_p^2 = \frac{a\mu_p^2 - 2b\mu_p + c}{ac - b^2}$$

We know that $A = \frac{a}{ac-b^2} = 4/9$, Hence we have $\sigma^2 = 4/9(\mu - 5/4)^2 + 1/8$

iii) Find the value of r_f

Answer: we have the mean returns of the tangency portfolio. Let's recover the Sharpe ratio (tangency gradient) of the CML and find the CML formula to recover r_f

$\sigma^2 = 4/9(\mu - 5/4)^2 + 1/8$

$\mu = \pm\sqrt{\frac{\sigma^2 - 1/8}{4/9}} + 5/4$

$\mu_{efficient} = 3/2\sqrt{\sigma^2 - 1/8} + 5/4$ drop negative

$\frac{d\mu}{d\sigma} = \frac{3\sigma}{2\sqrt{\sigma^2 - \frac{1}{8}}}$

Our CML gradient follows $\frac{d\mu}{d\sigma}$, where $\sigma_m = \frac{\sqrt{3}}{4}$ from our part ii solutions. Hence $S = \frac{d\mu}{d\sigma} = \frac{3\sqrt{3}}{2}$
Since $\mu_{CML} = S\sigma_{CML} + r_f$ we have

$$13/8 = \frac{3\sqrt{3}}{2} \cdot \frac{\sqrt{3}}{4} + r_f$$

Thus $r_f = \frac{1}{2}$
Otherwise our close form formula follows $\mu_m = \frac{c - br_f}{b - ar_f}$. Which we mentioned in the previous tutorial and u can recover r_f from this relationship

iv) Find the weight vector and variance of the market portfolio m

Answer: use our relationships and just solve

$$w_m = \frac{C^{-1}(\mu - r_f 1)}{1^T C^{-1}(\mu - r_f 1)} = (1/12, 7/12, 1/3)^T$$

$$\sigma_m^2 = 4/9(\mu_m - 5/4)^2 + 1/8 = 3/16$$

v) Find the equation of the capital market line. Give your answer in the form $\mu = f(\sigma)$

Answer: *refer to part iii

vi) Find the weight vector and mean of the portfolio d

Answer: Find the mean using simple algebraic methods and then use the 2 fund portfolio theorem to derive w_d

$$\mu_d = \mu_g - (\mu_m - \mu_g) = 2\mu_g - \mu_m = 5/2 - 13/8 = 7/8$$

Using 2 fund portfolio theorem we have

$$w_d = \alpha w_g + (1 - \alpha) w_m$$

Thus

$$\mu_d = \alpha\mu_g + (1 - \alpha)\mu_m \implies 7/8 = \frac{5}{4}\alpha + (1 - \alpha) \cdot 13/8$$

We have $\alpha = 2$, Thus $w_d = 2w_g - 1w_m = (5/12, -1/3, 11/12)^T$

vii) Find the correlation coefficient of portfolios m and d

Answer:

$$\begin{aligned}
\rho_{m,d} &= \frac{Cov(r_m, r_d)}{\sigma_m \sigma_d} \\
&= \frac{1}{\sigma_m^2} Cov(r_m, \alpha r_g + (1-\alpha) r_m) \\
&= \frac{1}{\sigma_m^2} \left[Cov(r_m, \alpha r_g) + Cov(r_m, (1-\alpha) r_m) \right] \\
&= \frac{1}{\sigma_m^2} \left[\alpha Cov(r_m, r_g) + (1-\alpha) Cov(r_m, r_m) \right] \\
&= \frac{1}{\sigma_m^2} \left[2\sigma_g^2 - \sigma_m^2 \right] \\
&= \frac{2\sigma_g^2 - \sigma_m^2}{\sigma_m^2} = 1/3
\end{aligned}$$

Question 3

Given the minimum variance frontier of risky assets:

$$\sigma^2 = \frac{a\mu^2 - 2b\mu + c}{ac - b^2}$$

and the Capital Market Line (CML):

$$\sigma = \frac{|\mu - r_f|}{\sqrt{c - 2br_f + ar_f^2}}.$$

Finding the Intersection

Squaring the CML equation gives

$$\sigma^2 = \frac{(\mu - r_f)^2}{c - 2br_f + ar_f^2}.$$

Equating the two expressions for σ^2:

$$\frac{a\mu^2 - 2b\mu + c}{ac - b^2} = \frac{(\mu - r_f)^2}{c - 2br_f + ar_f^2}.$$

Cross-multiplying:

$$(a\mu^2 - 2b\mu + c)(c - 2br_f + ar_f^2) = (ac - b^2)(\mu - r_f)^2.$$

Expanding and collecting terms in μ:

$$0 = (ar_f - b)^2 \mu^2 + 2(ar_f - b)(c - br_f)\mu + (c - br_f)^2.$$

Factorising:

$$\left[(ar_f - b)\mu + (c - br_f) \right]^2 = 0.$$

Hence, assuming $b - ar_f \neq 0$,

$$\boxed{\mu_T = \frac{c - br_f}{b - ar_f}}.$$

Substituting μ_T into the CML:

$$\begin{aligned}\sigma_T &= \frac{|\mu_T - r_f|}{\sqrt{c - 2br_f + ar_f^2}} \\ &= \frac{\left|\frac{c-br_f}{b-ar_f} - r_f\right|}{\sqrt{c - 2br_f + ar_f^2}} \\ &= \frac{\left|\frac{c-2br_f+ar_f^2}{b-ar_f}\right|}{\sqrt{c - 2br_f + ar_f^2}} \\ &= \frac{\sqrt{c - 2br_f + ar_f^2}}{|b - ar_f|}.\end{aligned}$$

Therefore, the intersection point is

$$\boxed{(\sigma_T, \mu_T) = \left(\frac{\sqrt{c - 2br_f + ar_f^2}}{|b - ar_f|}, \frac{c - br_f}{b - ar_f}\right)}.$$

Economic Interpretation

The intersection represents the **tangency portfolio** provided $b - ar_f > 0$, so that the point lies on the upper efficient frontier.

At this point, the Capital Market Line is tangent to the efficient frontier of risky assets.The tangency portfolio maximises the Sharpe ratio:

$$\boxed{S_T = \frac{\mu_T - r_f}{\sigma_T} = \sqrt{c - 2br_f + ar_f^2}}.$$

Economically, the tangency portfolio provides the highest expected excess return per unit of total risk among portfolios consisting entirely of risky assets.

Under the assumptions of mean-variance portfolio theory, investors can combine the risk-free asset with the tangency portfolio to construct portfolios along the CML.

Risk-averse investors may allocate part of their wealth to the risk-free asset, while investors seeking higher expected returns may borrow at the risk-free rate to leverage the tangency portfolio.

Thus, the tangency portfolio is the optimal riskyportfolio for all mean-variance investors who share the same investment opportunity set and can borrow and lend at the risk-free rate.